

Diptadeep Sinha

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Portfolio | LeetCode | GeeksforGeeks

EDUCATION

Kalinga Institute of Industrial Technology (KIIT) <i>Bachelor of Technology, Computer Science Engineering (CGPA: 9.52)</i>	Bhubaneswar, India Jul 2023 – Present
Contai Public School <i>ISC (Class XII), Science (PCMB) – 97.5%, AIR 10</i>	West Bengal, India Jul 2022

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C
Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks, SDLC
Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3
Backend: Node.js, Express.js, REST APIs
Databases: SQL, MySQL, MongoDB
Tools & Platforms: Git, GitHub, Docker, Linux, CI/CD, Vercel, Render, AWS, Postman, VS Code
Problem Solving: Solved 400+ algorithmic problems (DP, Graphs, Trees, Greedy)

EXPERIENCE

Full Stack Developer Intern – BYAMN Live Demo	Jun 2026 – Aug 2026
- Built secure REST APIs with Zod validation, authentication, and rate limiting, validated via Postman - Improved backend scalability using MongoDB indexing, pagination, and cascade deletion for data consistency - Wrote shared React/TypeScript components reused across 3+ pages to cut duplicate frontend code	

PROJECTS

GlideWay – Smart Bus Booking Platform GitHub Demo MERN, Socket.io, Mapbox	
- Built a bus booking platform tested to handle 500+ concurrent users without seat-allocation conflicts - Secured REST APIs with JWT auth and role-based access for passengers, operators, and admins - Used Socket.io and Mapbox for live bus tracking, and wrote a seat-locking algorithm to prevent double-booking	
AeroVariance – Urban Air Quality Intelligence Platform GitHub Demo Next.js, MongoDB, XGBoost	
- Built an AI forecasting pipeline (Next.js, FastAPI, XGBoost) predicting AQI up to 72 hours ahead across a live network of 6 CAAQMS stations - Layered SHAP-based explainability onto every forecast, along with a what-if simulator for policy impact - Cut dashboard load time from 14.6s to 2.2s via compound MongoDB indexing, and shipped multilingual (English, Hindi, Bengali) advisories across 100,000+ searchable locations - Containerized the application with Docker and deployed the production service on Render	
Merchanta AI – AI Representation Optimizer GitHub Demo Next.js, Python, FastAPI, MongoDB	
- Built a tool that scores how often a merchant's products surface in AI shopping assistant responses - Simulated 100+ buyer queries against product catalogs to rank which listings AI agents recommend - Pulled merchant data via Shopify APIs into MongoDB, served through FastAPI and Express.js endpoints	
Accessible Learning Platform for Visually Impaired GitHub Demo MERN, Web Speech API	
- Built voice navigation with the Web Speech API so visually impaired users could browse course content hands-free - Added a teacher approval step before uploaded content goes live, with role-based access for teachers vs. students - Implemented face-recognition authentication and AI-generated lesson summaries for accessible learning	

ACHIEVEMENTS

Smart India Hackathon 2025 <i>Waitlisted for Final Round</i>	India 2025
- Built a model to predict collision risk between satellites and debris using public orbital tracking datasets - Added uncertainty estimates to risk predictions to flag genuinely risky vs. uncertain cases - Owned the pipeline from raw orbital data cleaning to a deployed prediction model	

CERTIFICATIONS

AWS Academy Graduate – Cloud Foundations <i>Amazon Web Services (AWS) Training and Certification</i>	Credential
Cisco CCNA – Enterprise Networking, Security & Automation <i>Cisco Networking Academy</i>	Credential
IBM DevOps & Software Engineering Courses <i>Coursera</i>	Certificates